



THE LONDON SCHOOL
OF ECONOMICS AND
POLITICAL SCIENCE ■

HP407 – Evidence Review and Synthesis for Decision Making

SUMMATIVE SUBMISSION TEMPLATE 2017/18

CANDIDATE NUMBER: <i>(Five digit number available via LSE for You – this is not the same as the number on your LSE ID card!)</i>	<table><tr><td>8</td><td>8</td><td>9</td><td>6</td><td>4</td></tr></table>	8	8	9	6	4
8	8	9	6	4		
ASSESSMENT TITLE: <i>(E.g. Title of essay(s) attempted)</i>	EFFICACY OF OPIOID ANALGESICS FOR CHRONIC LOWER BACK PAIN: A SYSTEMATIC REVIEW AND META-ANALYSIS					
WORD COUNT: <i>Excludes references and appendices.</i>	3,083					

Please tick this box if you would be happy for your assessment to be shared with future cohorts: ☒

EFFICACY OF OPIOID ANALGESICS FOR CHRONIC LOWER BACK PAIN: A SYSTEMATIC REVIEW AND META-ANALYSIS

ABSTRACT

Background: Opioid analgesics are commonly used for lower back pain. The recent opioid crises in North America has incentivized rigorous research in recent years to inform an update of a review first published in 2016.

Objectives: To evaluate the effectiveness of opioid analgesics for chronic lower back pain.

Design: Three reviewers screened title and abstracts. Two reviewers assessed risk of bias and extracted data while a third acted as an arbitrator in case of disagreement.

Data Sources: CENTRAL (March 2018), MEDLINE (September 2015-March 2018), EMBASE (September 2015-March 2018), and PsycINFO (September 2015- March 2018).

Eligibility Criteria: Placebo-controlled RCTs in English and French of opioid analgesics in adults over the age of 18 with chronic lower-back pain measured by VAS or NRS pain-scales were included. Mean differences over 20 are considered clinically important.

Results: The pooled estimate on all RCTs using a random-effects model is a mean difference of -8.07 (95% CI:-8.71,-7.44). Overall heterogeneity is high ($I^2=65.3\%$) indicating variations in true-effect size across studies. Nevertheless, average drop-out rates were at 46.1% providing reason to believe that RCTs have low external validity. Most studies with an enrichment design (41.6%) show that patients receiving placebo after randomization go through withdrawal symptoms. 92% of studies had clear sponsorship from pharmaceutical companies and 73% of trial-arms had $p<0.01$.

Conclusion: Opioids provide non-clinically important changes in chronic lower back-pain in the short-term and intermediate-term.

1. INTRODUCTION

1.1 Rationale

Lower Back Pain (LBP) is the leading cause of disability worldwide.¹ LBP is characterized by tension or weakness in the legs caused by stresses on the muscles and ligaments that support the spine. Chronic LBP is deep, aching, or dull pain traveling down the legs lasting more than three months. Being one of the most common reasons for missed work, it poses serious economic burdens on families, governments, industries, and healthcare systems globally¹.

A systematic review for treatments of LBP showed that opioids account for 3-66% of LBP prescriptions². Over a ten-year period, Deyo et al. (2009) shows a 423% increase of Medicare expenditures for opioid prescriptions for LBP³ in the United States (US) and the Australian government noted a 15-fold increase in opioids dispensed for chronic non-cancer pain between 1992 and 2012⁴.

Current guidelines in many developed countries encourage the use of simple analgesics before the use of opioids^{5,6,7}. This is partly due to the growing recognition of the evidence that suggests opioids provide non-clinically significant improvement in pain as well as the role prescription opioids play in propagating the opioid epidemic in North America. The Center for Disease Control in the US reported that 40% of all opioid-related deaths could be attributed to prescription opioids in outpatient settings⁸. Nevertheless, opioids are still prescribed. In the US, findings show that 50% of those treated with opioids have chronic LBP⁹. In Australia, the three most common drugs for LBP are tramadol(8.2%), oxycodone(11.7%), and paracetamol and codeine combination(12.1%)^{10,11}. Understanding the effectiveness of opioids in LBP and their associated misuse is a health and public policy concern.

A systematic review by Shaheed et al. (2016) assessed the efficacy of opioid analgesics for lower back pain¹². The review included 20 randomized controlled trials (RCTs) that found a mean difference of 10.1 in short-term pain relief scores favoring opioids and a mean difference of 8.1 in intermediate-term pain relief scores favoring opioids¹². Opioids were effective but not clinically significant¹². Since 2016, more studies have identified the increasing rates of opioid misuse and deaths to guide novel developments of opioid drugs^{13,14}. For example, Rauck et al. (2015) found statistically significant differences in pain relief favoring a new combination of oxycodone with naltrexone versus placebo ($p=0.0004$) as well as an ability to minimize withdrawal using an extended-release formula¹³. Considering the abundance of new research on opioid effectiveness along with the added emphasis on opioid misuse, an update of the 2016 review is called for.

1.2 Objectives

To assess the effectiveness of opioid analgesics compared to placebo for chronic lower back pain.

2. METHODS

2.1 Eligibility Criteria

2.1.1 Inclusion Criteria

Population: Studies included adults aged 18 years and above with chronic LBP. LBP had to be gauged by the clinician and explicitly stated by the authors.

Intervention: Opioids included oxymorphone, morphine, transdermal fentanyl, oxycodone monotherapy, oxycodone/naloxone, tramadol monotherapy, tramadol/paracetamol, tapentadol, hydromorphone, naltrexone, transdermal buprenorphine, hydrocodone¹². We included brand and generic names for the opioid drugs described above to ensure we captured all reference to opioids in the literature.

Comparator: Placebo.

Outcomes: *Primary:* Pain is our primary outcome measure as it is why opioids are prescribed for LBP. We included studies that used the visual analogue scale (VAS) scores (range, 0-100) and the numerical rating scale (NRS) score (range, 0-10) to measure pain. These scales are used as they are common measurements in most chronic pain¹⁵ and back pain literature¹⁶. The NRS can easily be converted to the same 0-to-100 scale as the VAS because these two pain measures are highly correlated and can be used interchangeably¹⁷. *Secondary:* 1) use of enrichment design; 2) drop-out rates; 3) adverse event; 4) prevalence of opioid misuse as defined by the *The Diagnostic and Statistical Manual of Mental Disorders V* (DSM-5) or by the authors¹⁸. DSM-5 is a globally recognized standard for mental disorders and can be used as an objective tool to assess opioid misuse in RCTs.

Study Design: Included studies were RCTs of any duration that randomly allocated patients with LBP to an opioid analgesic or placebo. RCTs were chosen because they are the most rigorously conducted studies with the lowest threat to internal validity. No preference was given to enrichment versus non-enrichment studies to assess efficacy on a wider range of participants enrolled in RCTs to receive opioid analgesics. The search was restricted to October 2015-March 2018, based on updating Shaheed et al.'s paper¹².

2.1.2 Exclusion Criteria

We excluded population in the in-patient setting as this is more reflective of the type of environment where opioid use disorder can develop, according to the Center for Disease Control, and allows for patient characteristics to remain comparable across studies⁶. We did not exclude on the basis of co-morbidities or co-mediations to minimize this review's threats to external validity. We excluded studies that compared opioids with non-placebo comparators, other systematic reviews, and clinical observations.

2.2 Information Sources

We searched the following databases for placebo-controlled RCTs in English and French:

1. The Cochrane Central Register of Controlled Trials (searched 14 March 2018);
2. MEDLINE (via Ovid) (October 2015-March 14, 2018);
3. Embase (via Ovid) (October 2015-March 14, 2018);
4. PsycInfo (via Ovid) (October 2015- March 14, 2018).

2.3 Search

An example of a search strategy can be found in Appendix A.

2.4 Study Selection

Three reviewers divided studies evenly to screen titles and abstracts. Two review authors independently read full-text RCTs and reached agreement on inclusion by discussion. No studies in French were identified. Disagreements were solved with the third reviewer acting as an arbitrator.

2.5 Data Extraction

Two authors extracted data independently and checked for agreement where the third author acted as an arbitrator (Appendix B). We included information about the total number of participants treated, number of participants in each arm of the trial, study setting (location), demographic and clinical characteristics of the study samples (age, gender, pain baseline), opioid used and dosing regimen, study duration and follow-up, pain scales measured in mean differences and standard deviations, drop-out rates, and adverse events. We created a separate data extraction form for signs of misuse defined by the DSM-5 or the authors¹⁸.

2.6 Data Items

We considered pain relief (with respect to follow-up) in the short-term <3months, intermediate-term >=3 months<12 months, and long-term>12 months^{15,16}. When many follow-up-points were available for a time-

period, dates nearest to six weeks was chosen for short-term, six months for intermediate, and twelve for long-term. Based on LBP literature that categorizes time frames similarly, we considered this to be appropriate^{15,16}. It is possible that results could be interpreted differently if different time frames are used.

VAS and NRS pain scales were used as primary outcome measurements for this review; however, RCTs could be using other forms of pain-outcome measurement that are neglected by this inclusion criteria. We use the VAS as our baseline pain scale since it is used to categorize ‘clinically important’ differences in pain relief¹². According to LBP¹⁶ and chronic pain¹⁵ literature, a 20-point difference is a clinically important change in pain. Scores less than 10 mean pain change is unlikely to be noticed by patients¹².

2.7 Risk of Bias in individual studies

Two review authors independently assessed risk of bias for each study, using the criteria outlined in the Cochrane Handbook Table 8.7a¹⁹. Domains assessed under risk of bias can be found in Appendix C. Since four full-text RCTs were inaccessible, we could not perform a risk of bias assessment using Cochrane guidelines; however, Shaheed et al. ranked these studies based on the PEDro scale¹².

2.8 Summary Measures

Principal summary measures used were mean differences, standard deviation, and number of participants in each arm of the trial since the type of outcome data included from RCTs are continuous outcomes based on the interchangeable VAS and NRS pain scales.

2.9 Synthesis of Results

Data was extracted on a form created on Excel. After assessing all the studies qualitatively, 22 were deemed sufficiently homogenous to perform a meta-analysis using STATA. Pooled effects were calculated using a random-effects model because the different opioids used across RCTs had the potential to cause the true-effect sizes of the RCTs to vary from study to study.

As our baseline analysis, we divided the pooled effects into short-term and intermediate-term pain-relief to ensure results remained comparable. No additional sub-group analyses were conducted to prevent misleading interpretations and decrease the likelihood of false-positives and false-negatives¹⁹. To explore possible heterogeneity on interactions with effect-size, we performed a meta-regression on the following covariates¹⁹: 1) mode of administration, 2) log-transformed morphine equivalent dose-response, and 3) study design (i.e. enrichment vs. non-enrichment). The first could have clinically important implications on effectiveness

considering whether the opioid is short-acting or long-acting²⁰. The second was a post-hoc assessment after data-extraction showed variability in mode of administration and logs were deemed appropriate to deal with the skewedness of dosages²¹. The third has been controversial in regulatory settings as enrichment designs are thought to improve drug-efficacy data by screening out non-responsive patients²².

2.10 Risk of Bias across studies

Risks of bias that could affect the cumulative evidence considered publication bias and industry bias.

3. Results

3.1 Study Selection

1432 articles were identified from the databases (Figure 1). After deleting duplicates, there were 1105 articles. After titles and abstracts, we had 38 full text-articles to screen. Articles were excluded because ineligible study design, ineligible intervention or ineligible comparator. Two studies from Shaheed et al.'s systematic review were excluded as they did not use placebo comparators^{23,24}. We did not need to excluded studies for not reporting VAS or NRS pain scales as all studies reported outcomes in one of these two ways.

Four full-text articles from the Shaheed et al. systematic review could not be accessed^{25,26,27,28}; librarians from the London School of Economics and Political Science were asked to retrieve the inaccessible articles and were not able to. Data from these four full-text articles was still available from Shaheed et al. and this was included. We were left with 24 RCTs for a qualitative analysis and 22 for a meta-analysis of which 7 RCTs are new.

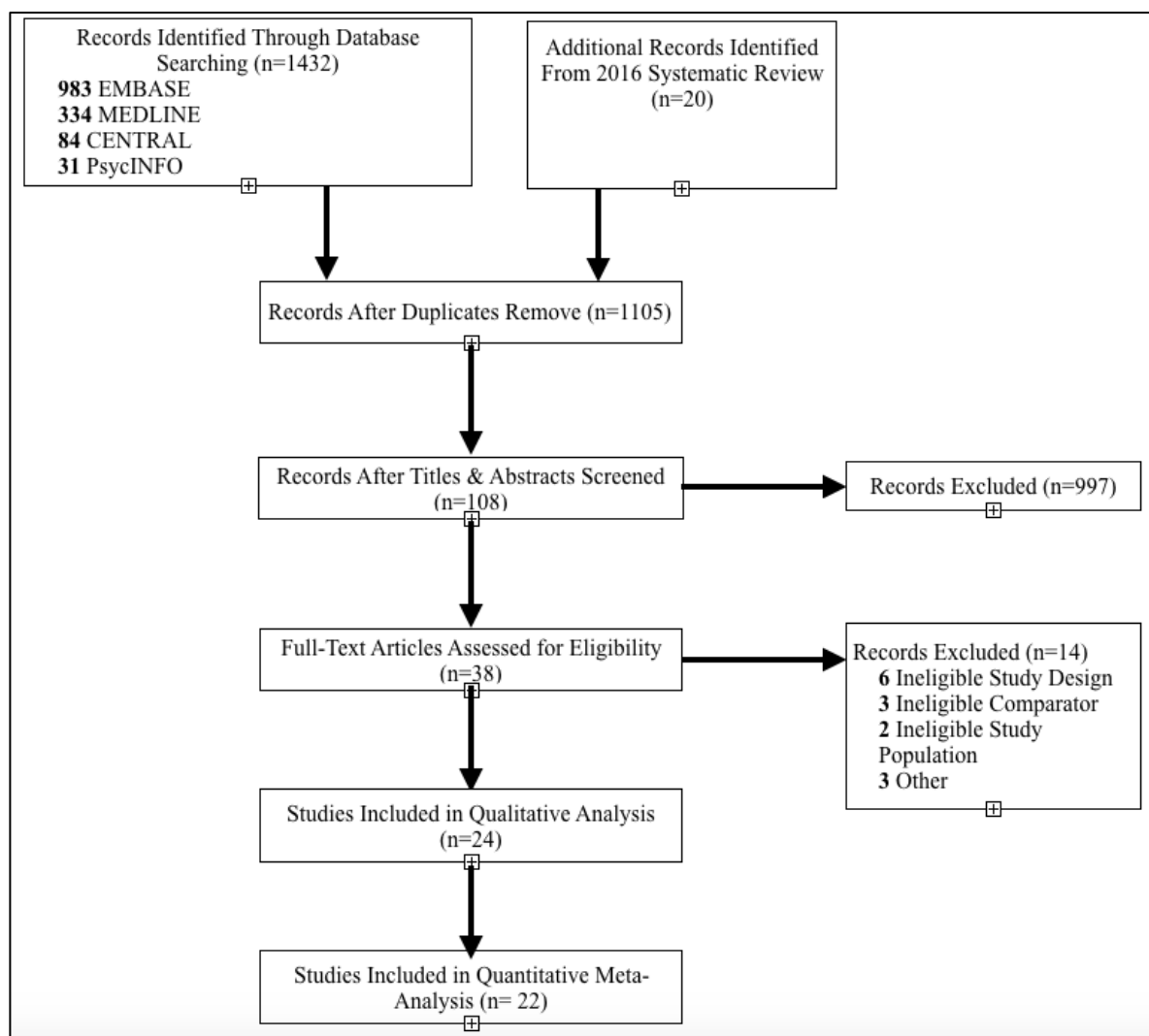


Figure 1. PRISMA-P Flow Chart

3.2 Characteristics of Included Studies

A detailed description of characteristics of studies included in this review is presented in Table 1. This review includes 8,883 participants with an average age of 49.94 years. 79% of the studies were conducted in the US and all studies were multi-centered. Tramadol was the most common opioid appearing in 24% of trial-arms and oxycodone was the second most common at 20%. No studies looked at long-term pain relief.

The most prevalent adverse event was constipation and withdrawal syndrome was recorded in 41.6% of RCTs. 75% of studies were enrichment design, meaning there was a run-in phase before randomization. Most studies with enrichment design recorded withdrawal in patients who were given placebo during randomization and dropped-out. Average drop-out rate across all RCTs was 46.1%. Other signs of opioid misuse included opioids being taken in larger amounts than intended, tolerance, and strong cravings. Two studies were not included in the quantitative analysis because they did not report standard deviations resulting in inadequate information for inclusion into the meta-analysis^{25,29}.

Table 1. Characteristics of Included Studies

Source	Patient Population Baseline	Mean Pain at Baseline (VAS)	Setting	Intervention	Morphine Equivalent Dosage (mg/day)	Comparison	Length of Treatment	Follow-up	Outcome Measure(s)	Industry Funding
Buynak et al, 2010 ^{**26}	965 Patients with chronic LBP; 406 M, 559 F, mean age: 49.9 y	n/a*	85 US centers; 15 CAN; 3 AUS	Tapentadol (313.2mg/d), N=313; Oxycodone MR (53mg/d), N = 322	125.28; 79.5	Placebo, N=313	12 wk	1, 8, 12 wk	NRS	Yes
Chu et al, 2012 ^{**25}	139 Patients with chronic LBP; 78 M, 61 F, mean age: 41.9 y	n/a	Clinical centers, Stanford	Morphine (78 mg/d) N=48	78	Placebo, N=55	1 mo	1 mo	VAS	Unclear
Cloutier et al, 2013 ³⁰	83 Patients with chronic LBP; 39 M, 44 F, mean age: 51.3 y	61.4 +/- 16.3	10 CAN centers	Oxycodone/naloxone (36.5mg/d) N=32	54.8	Placebo, N=31	8 wk	Week 8	VAS	Yes (Purdue Pharma)
Cristoph et al, 2017 ³¹	635 Patients with chronic LBP; 223 M, 412 F, mean age: 57.5	65.3 +/- 10.74	79 sites in 11 European countries	Cebranopadol (200hg) , N=129; Cebranopadol (400hg) , N=127, Cebranopadol (600hg) , N=127; Tapenadol (200mg) , N=126	n/a	Placebo, N=126	14 wk	Week 1, Week 2, Every 2 Weeks	NRS	Yes (Grunenthal GmbH)
Gimbel et al, 2016 ³²	511 patients with chronic LBP; 278 F, 233M, mean age: 53.0	62.8 +/- 12.65	66 Centers in the US	Buccal buprenorphine (300-1800 uh/d), N= 254	18	Placebo, N=257	12 wk	Week 12	NRS	Yes (Endo Pharmaceuticals)
Gordon et al, 2010 ³³	79 Patients with chronic LBP; 31 M, 47 F (1 unclear), mean age: 50.7	62.1 +/- 15.5	13 CAN centers	Transdermal buprenorphine (29.8ug/h) N=37	252	Placebo, N=42	8 wk	8 wk	VAS	Yes (Purdue Pharma)
Hale et al, 2010 ^{**28}	266 Patients with chronic LBP; 132 M, 134 F, mean age: 48.6 y	n/a**	66 US centers	Hydromorphone (26mg/d) N=133	104	Placebo, N=133	12 wk	1, 8, 12 wk	NRS	Yes (Endo Pharmaceuticals)

Hale et al, 2005 ³⁴	329 Patients with chronic LBP; 174 M, 155 F, mean age: 48.6 y	n/a*	Clinical centers, US	Oxymorphone MR (79.4 mg/d), N= 71 or Oxycodone MR , (155 mg/d), N=75	232.5; 238.2	Placebo, N= 67	18d	Days 7 and 18	VAS	Yes (Endo Pharmaceuticals)
Hale et al, 2007 ³⁵	142 Patients with chronic LBP; 78 M, 64 F, mean age: 47.1 y	69.5 +/- 16.5	30 centers in the US	Oxymorphone (OPANA) (80.9mg/d), N=70	242.7	Placebo, N=72	3 mo	Day 10, <3mo	VAS	Yes (Endo Pharmaceuticals)
Hale et al, 2015 ³⁶	370 opioid-experienced and opioid-naïve patients with chronic LBP >= 3months; 182 M, 189 F, mean age 51.8	n/a	78 sites in US	Hydrocodone (48mg/day), N= 191	48	Placebo, N= 180	12 weeks	Weeks 0 and 12	NRS	Yes (Teva Branded Pharmaceuticals)
Katz et al, 2007 ^{27**}	325 Patients with chronic LBP; 126 M, 109 F, mean age: 49.8 y	n/a	Multicenter US	Oxymorphone ER (39.2 mg/d) N=105	117.6	Placebo, N=100	12 wk	Weeks 1 and 8, 90d	VAS	Yes (Endo Pharmaceuticals)
Katz et al, 2015 ¹⁴	389 Patients with chronic LBP; 184 M, 205 F; mean age: 51	64.26	46 Centers in the US	Xamptza (Extended-Release Oxycodone) (60mg/d) N=193	90	Placebo, N= 196	12 Wk	Week 6, Week 12	NRS	Yes (Collegium Pharmaceuticals)
Peloso et al, 2004 ³⁷	336 Patients with chronic LBP; 126 M, 210 F, mean age: 57.5 y	67.9 +/- 14.95	US	Tramadol/paracetamol (158mg/d) N=167	15.8	Placebo, N= 169	91 d	91d	VAS	Yes (Ortho-McNeil Pharmaceuticals)
Rauck et al, 2014 ³⁸	510 Patients with chronic LBP; 235 M, 275 F, mean age: 49.0 y	64.26 +/- 12.76	59 Sites in the US	Hydrocodone (119 mg/d) N=151	119	Placebo, N=151	12 wk	Day 85	NRS	Yes (Zogenix)
Rauck et al, 2015 ¹³	281 Patients with chronic LBP; mean age: 50.0	64.26	47 centers in the US	ALO-02 (oxycodone extended-release/naloxone) , (70mg/d), N= 147	105	Placebo, N=134	12 wk	Weekly	NRS	Yes (Pfizer)
Rauck et al, 2016 ³⁹	461 Patients with chronic LBP; 259 F, 202 M, NRS>4.0, mean age: 50.7	66.3 +/- 10.0	60 centers in the US	Buccal buprenorphine (600uh/d), N=229	18	Placebo, N=232	12 wk	Week 12	NRS	Yes (Endo Pharmaceuticals)
Ruoff et al, 2003 ⁴⁰	318 Patients with chronic LBP; 235 M, 275 F, mean age: 49.0 y	69.95 +/- 14.6	29 Centers in the US	Tramadol/paracetamol (157.5 mg/d), N= 91	15.75	Placebo N=74	91 d	91 d	VAS	Yes (Ortho-McNeil Pharmaceuticals)

Schiporst et al, 2013 ²⁹	50 Participants with chronic LBP; 16 M, 34 F, mean age: 43.0 y	49.6 +/- 39.9	2 centers in the Netherlands	Tramadol/paracetamol , N= 25	n/a	Placebo N=25	14d	14d	VAS	Crunenthal BV and Stichting Beatrixwoord, The Netherlands
Schnitzer et al, 2000 ²⁹	254 Patients with chronic LBP; 127 M, 127 F, mean age: 47.1 y	65.6 +/- 15.4	26 Centers in the US	Tramadol (242mg/d) N=127	24.2	Placebo, N= 127	4wk	49 d	VAS	Yes (Ortho-McNeil Pharmaceuticals)
Steiner et al, 2011 ²⁴	541Patients with chronic LBP; 242 M, 298 F, mean age: 49.4 y	66.1 +/- 11.56	86 Centers in the US	Transdermal Buprenorphine , (15ug/h) N= 257	189	Placebo, N=283	84d	8 wk	NRS	Yes (Purdue Pharma)
Uberall et al, 2012 ⁴¹	355 Patients with chronic LBP; 135 M, 220 F, mean age: 58.5 y	53.24 +/- 11.5	31 Sites in Germany	Transdermal buprenorphine (10 ug/h) N=107	126	Placebo, N=110	6wk	4 wk	NRS	Yes (TEVA Germany)
Vorsanger et al, 2008 ⁴²	386 Patients with chronic LBP, 192 M, 194 F, mean age: 47.8 y	71.2	30 Centers in the US	Tramadol ER (300 mg/d) N=127; Tramadol ER (200 mg/d) , N=129	30; 20	Placebo, N=126	12 wk	1, 8, 12 wk	VAS	Yes (Biovail Corp)
Webster et al, 2006 ⁴³	719 Patients with chronic LBP; 277 M, 442 F, mean age: 48.1y	69.39 +/- 12.59	45 Sites in the US	Oxycodone (34.5mg/d) plus low dose naltrexone four times daily; N= 206; Oxycodone (39mg/d) N=206; Oxycodone/Naltrexone (34.7 mg/d) plus low dose naltrexone twice daily; N=206	58.5; 51.75; 52.05	Placebo N=101	12 wk	Weeks 1 and 12	VAS	Unclear
Wen et al, 2015 ⁴⁴	439 Patients with chronic LB; 121 M 191 F, mean age: 47.3	66.09 +/- 10.74	76 sites in the US	Hydrocodone tablets (50mg/d); N= 229	50	Placebo, N=210	12 wk	Week 12	NRS	Yes (Purdue Pharma)

*n/a= data not available from study **articles inaccessible, data extracted from Shaheed et al. (2016) ***Morphine equivalent dosages calculated using tool provided by described by State of Ohio Board of Pharmacy⁴⁵

3.3 Risk of bias within studies

For a detailed overview of each study refer to Appendix C. Risk of bias assessments suggests RCTs control for internal validity regarding blinding, but was unclear with respect to some sequence generation, selection bias and selective outcome reporting since protocols could not be retrieved. Incomplete outcome data had high risk of bias since studies had high drop-out rates and excluded those patients from final results.

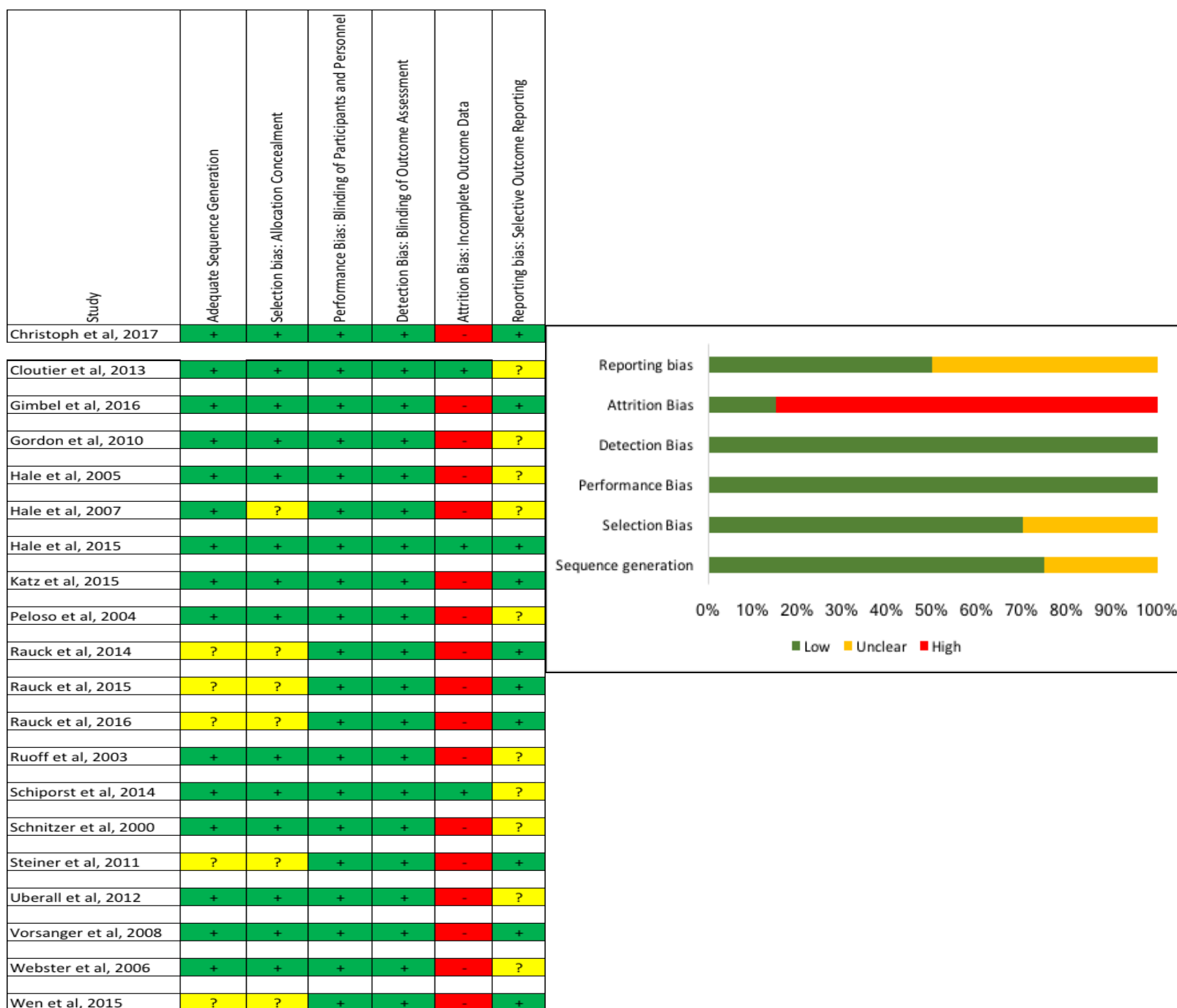


Figure 2. Summary of Risk of Bias

3.4 Results of Individual Studies

The pooled estimate of all RCTs using a random-effects model is a mean difference (MD) of -8.07 (95% CI:-8.71,-7.44). If statistical significance is taken to be $p < 0.10$, overall heterogeneity is substantial ($I^2 = 65.3\%$) and significant ($p = 0.000$), supporting our initial hypothesis of variations in true-effect sizes across studies¹⁹. Results suggest there is evidence that opioids are effective compared to placebo for the treatment of LBP.

Pooled estimates on short-term pain-relief is -8.45 (95% CI:-9.74,-7.16) and on intermediate pain-relief is -7.96 (95% CI:-8.68,-7.23). Confidence intervals suggest that there is no tangible difference between short-term and intermediate-term pain relief or across studies. Within short-term and intermediate-term heterogeneity are substantial and significant ($I^2 = 71.4\%$, 59.1% respectively, $p = 0.000$)¹⁹. Heterogeneity between groups is not statistically significant ($p = 0.515$).

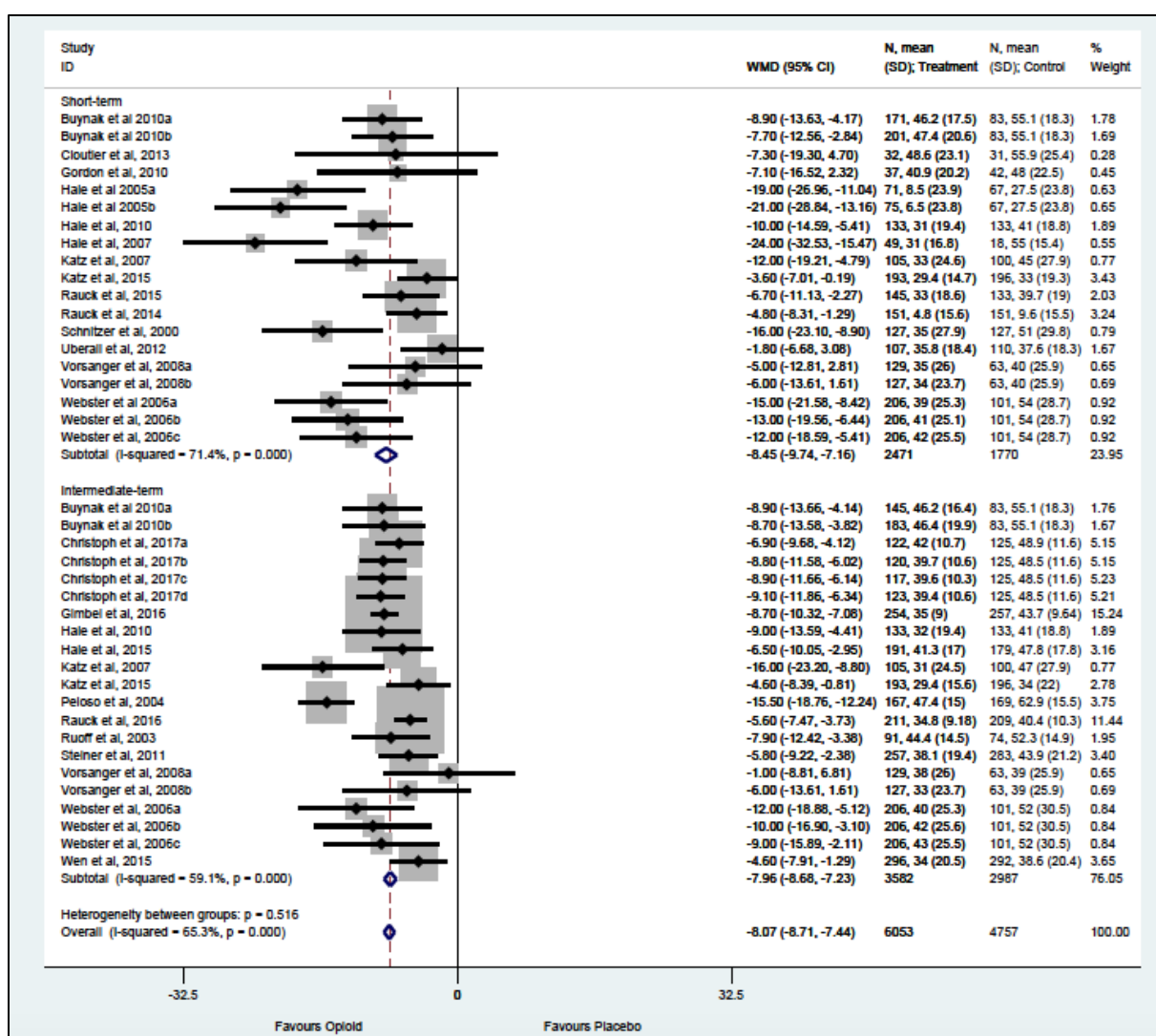


Figure 3. Forest plot for opioid effective on short-term and intermediate-term pain relief compared to placebo.

3.5 Risk of Bias Across Studies

The funnel-plot generated for this review is relatively symmetric (Figure 4a). A funnel-plot with contour lines (Figure 4b) shows 27 trial-arms were extremely significant ($p < 0.01$), 4 were statistically significant, ($0.01 < p < 0.05$), and 6 were not statistically significant ($p > 0.1$).

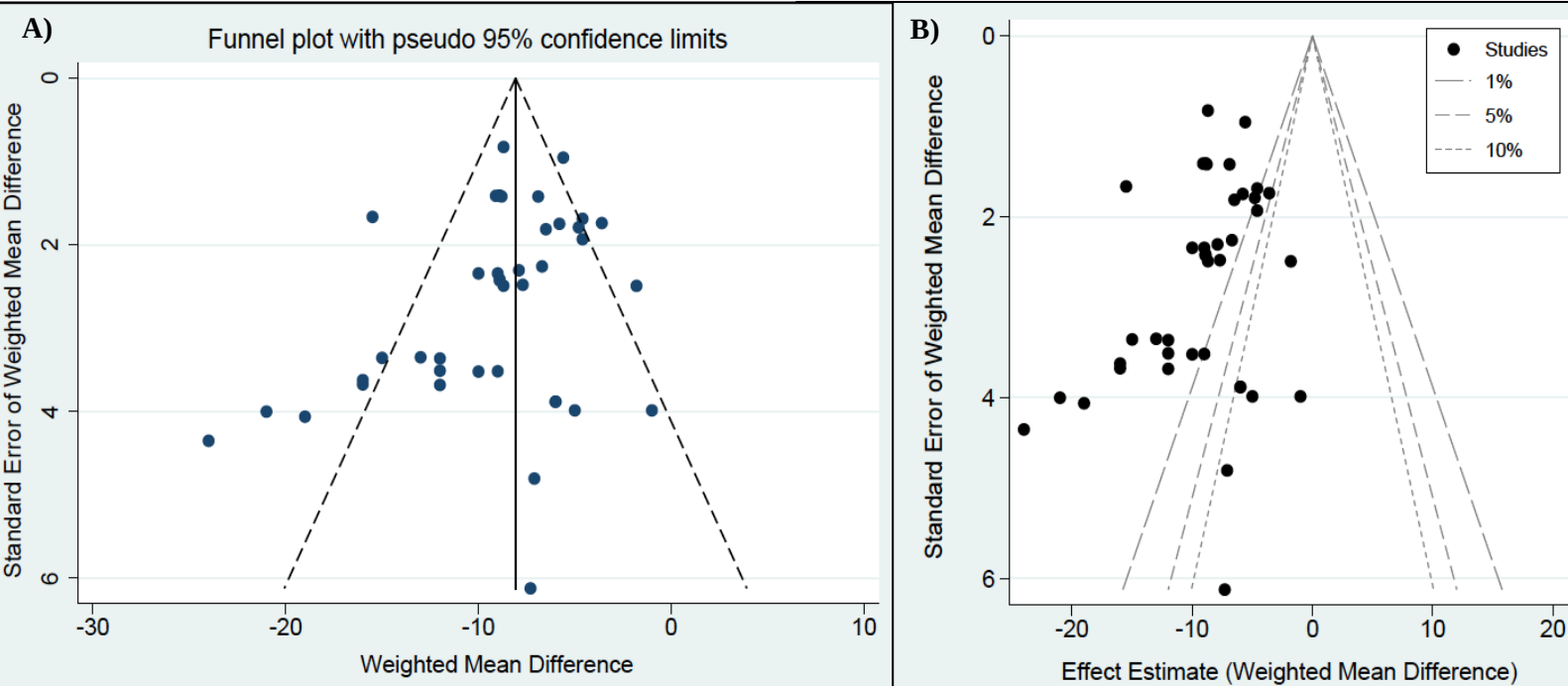


Figure 4. a) Funnel plot b) Funnel plot with contour line

3.6 Meta-regressions

Meta-regressions were conducted to evaluate the interaction of treatment effects and potential covariates (Graphs, Appendix D). Results suggest that study design (enrichment vs. non-enrichment) is not a statistically significant explanatory variable in treatment effect ($R^2=0.0879$, $p=0.287$). Method of administration (i.e. oral, transdermal, buccal) is not a statistically significant explanatory variable in treatment effect ($R^2=0.014$, $p=0.171$). Log-transformed morphine equivalent dose-response is also not statistically significant ($p=0.488$), where the relative change in treatment effect is a 1.5 decrease in pain per unit increase in morphine equivalent dose.

4. Discussion

4.1 Summary of Evidence

This meta-analysis shows that opioids are effective at reducing pain compared to placebo with an overall MD -8.07 (95% CI:-8.71,-7.44). Short-term pain relief shows non-significant effectiveness with MD -8.45 (95% CI:-9.74,-7.16) than intermediate pain-relief with MD -7.96 (95% CI:-8.68,-7.23). No studies assessed long-term pain relief. Nevertheless, these results should be interpreted with caution since clinically important changes in pain are at least a 20-point difference and changes less than 10 are not considered to be noticed by patients. It is important to note, however, that these results are aggregated MDs and on an individual level, patients may experience clinically meaningful difference in LBP relief⁴⁶. Considering this, opioids provide non-clinically important changes in chronic LBP. These results are consistent with Shaheed et al.'s (2016) review¹².

Overall heterogeneity ($I^2=65.3\%$) is high meaning that there was variability across studies. When considering the sub-groups of short-term pain relief ($I^2=71.4\%$) and intermediate-term pain relief ($I^2=59.1\%$), less of a difference is noticed across RCTs in the intermediate-term than in the short-term. Possible reasons for heterogeneity include clinical diversity¹⁹ such as the use of different morphine-equivalent dosages or different methods of administering the opioid across studies. Meta-regressions show dose-response is not a statistically significant explanation for heterogeneity.

Heterogeneity could also be explained by methodological diversity¹⁹ such as the type of study design used or risks of biases within individual studies, such as high attrition bias and unclear selective outcome reporting. A meta-regression shows no significant association between different study designs and treatment effect. Although RCTs pose a minimal threat to internal validity, readers should consider the risk of bias in these domains when interpreting results which could make published results misleading. In the funnel-plot with contour lines, there is a suggestion of missing non-statistically significant studies ($p>0.1$) where publication bias is a probable explanation¹⁹. This may be explained by the high influence of commercial interests observed in 92% of RCTs.

These results are consistent with current guidelines that emphasize the use of simple analgesics, or even yoga, before the prescription of opioids for chronic LBP^{5,6,7}; however, since opioids are still commonly prescribed, different enforcement incentives for physicians should be considered to cut-down unnecessary opioid prescriptions for chronic LBP. Another important consideration for physicians is that these results

are not generalizable; 46.1% of patients who started the study dropped-out due an incapacity for metabolizing the drugs properly²², side-effects, and withdrawal symptoms. This could mean a significant proportion of patients with chronic LBP in the real world may not be suitable for opioids yet are still prescribed them. This review also shows (Appendix B) that within 41.6% the studies, people who were randomized to placebo after run-in phase experienced signs of withdrawal. This could be propagating an opioid crisis in North America; however, more studies must be done to make this conclusive. Researchers and clinicians should continue using standardized subjective or clinical opioid withdrawal scales (SOWS or COWS), or use the DSM-5 as a tool to gauge patient's opioid misuse and abuse throughout a trial.

4.2 Limitations

This review did not include head-to-head RCTs (i.e. opioid vs. opioid) which could provide greater indirect evidence into the comparative effectiveness of opioids. Mean pain at baseline and morphine equivalent dosages could not be retrieved from some studies, which might affect the comparability of evidence and thus their inclusion into the meta-analysis. Standard deviations could not be retrieved for two studies and thus were not included in the meta-analysis; authors should be contacted for full retrieval of data of identified research. This review attempted to explore heterogeneity, however, meta-regressions were not explanatory and more research should be conducted on different variables to explain it.

4.3 Conclusions

This review shows that opioids provide non-clinically important changes in chronic LBP in the short-term and intermediate-term compared to placebo. This review adds the additional lens on assessing signs of opioid misuse in RCTs. Future research should consider the comparative effectiveness of opioids or the comparative effectiveness of opioids vs. simple analgesics in providing clinically important changes in LBP. More studies should be directed at understanding long-term effectiveness. Future research should also consistently monitor patients for signs of withdrawal and build on the literature in this way as it is an important public health and policy concern.

Works Cited

1. Hoy D, March L, Brooks P, et al. The global burden of low back pain: estimates from the Global Burden of Disease 2010 study. *Ann Rheum Dis*. 2014;73(6):968-974. doi:10.1136/annrheumdis-2013-204428.
2. Martell BA, O'Connor PG, Kerns RD, et al. Systematic review: opioid treatment for chronic back pain: prevalence, efficacy, and association with addiction. *Ann Intern Med*. 2007;146(2):116-127.
3. Deyo RA, Mirza SK, Turner JA, Martin BI. Overtreating chronic back pain: time to back off? *J Am Board Fam Med*. 2009;22(1):62-68. doi:10.3122/jabfm.2009.01.080102.
4. Hauser W, Schug S, Furlan AD. The opioid epidemic and national guidelines for opioid therapy for chronic noncancer pain: a perspective from different continents. *Pain reports*. 2017;2(3):e599. doi:10.1097/PR9.0000000000000599.
5. Busse JW, Juurlink DN, Buckley DN, Guyatt GH. The authors respond to "Inconsistencies in the 2017 Canadian Guideline for Opioids for Chronic Noncancer Pain". *CMAJ*. 2017;189(39):E1234-E1235. doi:10.1503/cmaj.733313.
6. Dowell D, Haegerich TM, Chou R. CDC Guideline for Prescribing Opioids for Chronic Pain--United States, 2016. *JAMA*. 2016;315(15):1624-1645. doi:10.1001/jama.2016.1464.
7. Newman R (RACGP). *Prescribing Drugs of Dependence in General Practice, Part A Clinical Governance Framework*.; 2015.
https://westvicphn.com.au/images/HOME/Health_Professionals/Alcohol_Other_Drugs/Addictive-drugs-guide-A.pdf.
8. Opioid overdoses in the United States. *J Pain Palliat Care Pharmacother*. 2012;26(1):44-47. doi:10.3109/15360288.2011.653875.
9. Hudson TJ, Edlund MJ, Steffick DE, Tripathi SP, Sullivan MD. Epidemiology of regular prescribed opioid use: results from a national, population-based survey. *J Pain Symptom Manage*. 2008;36(3):280-288. doi:10.1016/j.jpainsymman.2007.10.003.
10. Briggs AM, Buchbinder R. Back pain: a National Health Priority Area in Australia? *Med J Aust*. 2009;190(9):499-502.
11. Williams CM, Maher CG, Hancock MJ, et al. Low back pain and best practice care: A survey of general practice physicians. *Arch Intern Med*. 2010;170(3):271-277. doi:10.1001/archinternmed.2009.507.
12. C.A. S, C.G. M, K.A. W, et al. Efficacy, Tolerability, and Dose-Dependent Effects of Opioid Analgesics for Low Back Pain: A Systematic Review and Meta-analysis. *JAMA Intern Med*. 2016;176(7):958-968. doi:https://dx.doi.org/10.1001/jamainternmed.2016.1251.
13. R.L. R, M.E. H, A. B, et al. A randomized double-blind, placebo-controlled efficacy and safety study of ALO-02 (extended-release oxycodone surrounding sequestered naltrexone) for moderate-to-severe chronic low back pain treatment. *Pain*. 2015;156(9):1660-1669. doi:https://dx.doi.org/10.1097/j.pain.0000000000000230.
14. N. K, E.A. K, M. O, et al. A phase 3, multicenter, randomized, double-blind, placebo-controlled, safety, tolerability, and efficacy study of Xtampza ER in patients with moderate-to-severe chronic low back pain. Ashworth Birnbaum, Breitbart, Chaparro, Dart, Deuschle, Dworkin, Dworkin, Field, Fleming, Hale, Hart, Hedeker, Hogan, Jamison, Johannes, Katz, Katz, Katz, Katz, Katz, Katz, Kopecky, Leber, Mastropietro, Moore, Portenoy, Rauck, Rosenblum, Schnoll, S B, ed. *Pain*. 2015;156(12):2458-2467. doi:http://dx.doi.org/10.1097/j.pain.0000000000000315.
15. Dworkin RH, Turk DC, Wyrwich KW, et al. Interpreting the clinical importance of treatment outcomes in chronic pain clinical trials: IMMPACT recommendations. *J Pain*. 2008;9(2):105-

121. doi:10.1016/j.jpain.2007.09.005.
16. Ostelo RWJG, Deyo RA, Stratford P, et al. Interpreting change scores for pain and functional status in low back pain: towards international consensus regarding minimal important change. *Spine (Phila Pa 1976)*. 2008;33(1):90-94. doi:10.1097/BRS.0b013e31815e3a10.
17. Hawker GA, Mian S, Kendzerska T, French M. Measures of adult pain: Visual Analog Scale for Pain (VAS Pain), Numeric Rating Scale for Pain (NRS Pain), McGill Pain Questionnaire (MPQ), Short-Form McGill Pain Questionnaire (SF-MPQ), Chronic Pain Grade Scale (CPGS), Short Form-36 Bodily Pain Scale (SF. *Arthritis Care Res (Hoboken)*. 2011;63 Suppl 1:S240-52. doi:10.1002/acr.20543.
18. Association AP. *DSM 5*; 2013. doi:10.1176/appi.books.9780890425596.744053.
19. Higgins JP, Green S. *Cochrane Handbook for Systematic Reviews of*; 2008. doi:10.1002/9780470712184.
20. Argoff CE, Silvershein DI. A comparison of long- and short-acting opioids for the treatment of chronic noncancer pain: tailoring therapy to meet patient needs. *Mayo Clin Proc*. 2009;84(7):602-612. doi:10.1016/S0025-6196(11)60749-0.
21. Feng C, Wang H, Lu N, et al. Log-transformation and its implications for data analysis. *Shanghai Arch psychiatry*. 2014;26(2):105-109. doi:10.3969/j.issn.1002-0829.2014.02.009.
22. Campbell J, King NB. "Unsettling circularity": Clinical trial enrichment and the evidentiary politics of chronic pain. *Biosocieties*. 2017;12(2):191-216. doi:10.1057/biosoc.2016.7.
23. Allan L, Richarz U, Simpson K, Slappendel R. Transdermal fentanyl versus sustained release oral morphine in strong-opioid naïve patients with chronic low back pain. *Spine (Phila Pa 1976)*. 2005;30(22):2484-2490. doi:10.1097/01.brs.0000186860.23078.a8.
24. Steiner D, Munera C, Hale M, Ripa S, Landau C. Efficacy and safety of buprenorphine transdermal system (BTDS) for chronic moderate to severe low back pain: A randomized, double-blind study. *J Pain*. 2011;12(11):1163-1173. doi:10.1016/j.jpain.2011.06.003.
25. Chu LF, D'Arcy N, Brady C, et al. Analgesic tolerance without demonstrable opioid-induced hyperalgesia: A double-blinded, randomized, placebo-controlled trial of sustained-release morphine for treatment of chronic nonradicular low-back pain. *Pain*. 2012;153(8):1583-1592. doi:10.1016/j.pain.2012.02.028.
26. Buynak R, Shapiro DY, Okamoto A, et al. Efficacy and safety of tapentadol extended release for the management of chronic low back pain: results of a prospective, randomized, double-blind, placebo- and active-controlled Phase III study. *Expert Opin Pharmacother*. 2010;11(11):1787-1804. doi:10.1517/14656566.2010.497720.
27. Katz N, Rauck R, Ahdieh H, Ma T. A 12-week , randomized , placebo-controlled trial assessing the safety and efficacy of oxycodone extended release for opioid-naïve patients with chronic low back pain*. *Curr Med Res Opin*. 2007;23(1):117-128. doi:http://dx.doi.org/10.1185/030079906X162692.
28. Hale M, Khan A, Kutch M, Li S. Once-daily OROS hydromorphone ER compared with placebo in opioid-tolerant patients with chronic low back pain. *Curr Med Res Opin*. 2010;26(6):1505-1518. doi:10.1185/03007995.2010.484723.
29. Schiphorst Preuper HR, Geertzen JHB, Van Wijhe M, et al. Do analgesics improve functioning in patients with chronic low back pain? An explorative triple-blinded RCT. *Eur Spine J*. 2014;23(4):800-806. doi:10.1007/s00586-014-3229-7.
30. Cloutier C, Taliano J, O'Mahony W, et al. Controlled-release oxycodone and naloxone in the treatment of chronic low back pain: a placebo-controlled, randomized study. *Pain Res Manag*. 2013;18(2):75-82. doi:10.1155/2013/164609.

31. Christoph A, Eerdekens M-H, Kok M, Volkers G, Freynhagen R. Cebranopadol, a novel first-in-class analgesic drug candidate: first experience in patients with chronic low back pain in a randomized clinical trial. *Pain*. 2017;158(9):1813-1824. doi:<https://dx.doi.org/10.1097/j.pain.0000000000000986>.
32. Gimbel J, Spierings ELH, Katz N, et al. Efficacy and tolerability of buccal buprenorphine in opioid-experienced patients with moderate to severe chronic low back pain: results of a phase 3, enriched enrollment, randomized withdrawal study. *Pain*. 2016;157(11):2517-2526. doi:<http://dx.doi.org/10.1097/j.pain.0000000000000670>.
33. Gordon A, Rashedi S, Moulin DE, et al. Buprenorphine transdermal system for opioid therapy in patients with chronic low back pain. *Pain Res Manag*. 2010;15(3):169-178. doi:10.1155/2010/216725.
34. Hale ME, Dvergsten C, Gimbel J. Efficacy and safety of oxymorphone extended release in chronic low back pain: results of a randomized, double-blind, placebo- and active-controlled phase III study. *J Pain Off J Am Pain Soc*. 2005;6(1):21-28. doi:10.1016/j.jpain.2004.09.005.
35. Hale ME, Ahdieh H, Ma T, Rauck R. Efficacy and Safety of OPANA ER (Oxymorphone Extended Release) for Relief of Moderate to Severe Chronic Low Back Pain in Opioid-Experienced Patients: A 12-Week, Randomized, Double-blind, Placebo-controlled Study. *J Pain*. 2007;8(2):175-184. doi:10.1016/j.jpain.2006.09.011.
36. Hale ME, Laudadio C, Yang R, Narayana A, Malamut R. Efficacy and tolerability of a hydrocodone extended-release tablet formulated with abuse-deterrence technology for the treatment of moderate-to-severe chronic pain in patients with osteoarthritis or low back pain. *J Pain Res*. 2015;8:623-636. doi:<https://dx.doi.org/10.2147/JPR.S83930>.
37. Peloso PM, Fortin L, Beaulieu A, Kamin M, Rosenthal N, Protocol TRP-CAN-1 Study Group PT-C-1 S. Analgesic efficacy and safety of tramadol/ acetaminophen combination tablets (Ultracet) in treatment of chronic low back pain: a multicenter, outpatient, randomized, double blind, placebo controlled trial. *J Rheumatol*. 2004;31(12):2454-2463. <http://www.ncbi.nlm.nih.gov/pubmed/15570651>.
38. Rauck RL, Nalamachu S, Wild JE, et al. Single-entity hydrocodone extended-release capsules in opioid-tolerant subjects with moderate-to-severe chronic low back pain: A randomized double-blind, placebo-controlled study. *Pain Med (United States)*. 2014;15(6):975-985. doi:10.1111/pme.12377.
39. R.L. R, J. P, Q. X, et al. Efficacy and tolerability of buccal buprenorphine in opioid-naïve patients with moderate to severe chronic low back pain. *Postgrad Med*. 2016;128(1):1-11. doi:<https://dx.doi.org/10.1080/00325481.2016.1128307>.
40. Ruoff GE, Rosenthal N, Jordan D, Karim R, Kamin M. Tramadol/acetaminophen combination tablets for the treatment of chronic lower back pain: A multicenter, randomized, double-blind, placebo-controlled outpatient study. *Clin Ther*. 2003;25(4):1123-1141. doi:10.1016/S0149-2918(03)80071-1.
41. Uberall MA, Mueller-Schwefe GHH, Terhaag B. Efficacy and safety of flupirtine modified release for the management of moderate to severe chronic low back pain: results of SUPREME, a prospective randomized, double-blind, placebo- and active-controlled parallel-group phase IV study. *Curr Med Res Opin*. 2012;28(10):1617-1634. doi:10.1185/03007995.2012.726216.
42. Vorsanger GJ, Xiang J, Gana TJ, Pascual ML, Fleming RR. Extended-release tramadol (tramadol ER) in the treatment of chronic low back pain. *JOpioidManag*. 2008;4(2):87-97.
43. Webster LR, Butera PG, Moran L V., Wu N, Burns LH, Friedmann N. Oxytrex Minimizes Physical Dependence While Providing Effective Analgesia: A Randomized Controlled Trial in

- Low Back Pain. *J Pain*. 2006;7(12):937-946. doi:10.1016/j.jpain.2006.05.005.
44. Wen W, Sitar S, Lynch SY, He E, Ripa SR. A multicenter, randomized, double-blind, placebo-controlled trial to assess the efficacy and safety of single-entity, once-daily hydrocodone tablets in patients with uncontrolled moderate to severe chronic low back pain. *Expert Opin Pharmacother*. 2015;16(11):1593-1606. doi:<https://dx.doi.org/10.1517/14656566.2015.1060221>.
45. Schierholt SW, Kasich JR. Oral Morphine Milligram Equivalent Conversion Table. *State Ohio Board Pharm*. 2017;(614).
<https://www.ohiopmp.gov/Documents/MorphineEquivalentDailyDoseConversionTable.pdf>.
46. Olsen MF, Bjerre E, Hansen MD, et al. Pain relief that matters to patients: systematic review of empirical studies assessing the minimum clinically important difference in acute pain. *BMC Med*. 2017;15(1):35. doi:10.1186/s12916-016-0775-3.

Acknowledgements

All work was divided among group members and each author contributed to the project equally.

Appendix A- MEDLINE (Ovid) Example Search Strategy:

1. exp Analgesics, Opioid/
2. exp analgesics, opioid/ or exp alfentanil/ or exp alphaprodine/ or exp buprenorphine/ or exp buprenorphine, naloxone drug combination/ or exp butorphanol/ or exp codeine/ or exp dextromoramide/ or exp dextropropoxyphene/ or exp dihydromorphine/ or exp diphenoxylate/ or exp "encephalin, ala(2)-mephe(4)-gly(5)-/ or exp "encephalin, d-pencillamine (,25)-"/ or exp ethylketocyclozavine/ or exp ethylmorphine/ or exp etorphine/ or exp fentanyl/ or exp heroin/ or exp hydrocordone/ or exp levorphanol/ or exp meperidine/ or exp meptazinol/ or exp Oxymorphone/ or methadone/ or exp methadyl acetate/ or exp morphine/ or exp nalbuphine/ or exp opiate alkaloids/ or exp opium/ or exp oxycodone/ or exp pentazocine/ or exp phenazocine/ or exp phenperidine/ or exp pirinitramide/ or exp promedol/ or exp sufentanil/ or exp tillidine/ or exp tramadol
3. (alfentanil or alphaprodine or "bensomorphan adj25 deriv*" or beztiramide or buprenorphine or "buprenorphine, naloxone drug combination" or butorphanol or capodex or codeine or dextromoramide or dextropropoxyphene or dexocine or di-gesic or dihydrocodeine or dihydromorphine or dilaudid or diphenoxylate or diphenylpropylamine or durogesic or endone or ethylketocyclazocine or ethylmorphine or etorphine or Fentanyl or hydrocordone or hydromorphone or ketobemidone or levorphanol or meperidine or mepazinol or methadone or "methadyl acetate" or morphine or "morphinan adj25 deriv*" or nalbuphine or no2a* or norspan or OPANA\$ or "opi* adj25 alkaloid\$" or "opiate alkaloids" or opium or ordine or "oripaviin adj25 deriv*" or oxy* or oxycodone or Oxymorphone or pentazocine or pethidine or phenazocine or phenoperidine or "phenylpiperidine adj25 deriv*" or physeptone or pirinitramide or piritramide or promedol or suboxone or subutex or sulfentanil or tapendatol or tapendol or targin or tilidine or tramadol).mp.
4. opioid*.mp.
5. 1 or 2 or 3 or 4
6. Low Back Pain/
7. ("low\$ back pain" or "low\$ back ach\$" or lumbago or "lor\$ adj25 back pain" or "low\$ adj25 back ach\$" or "low\$ back strain" or "low\$ back pain\$" or simpl\$ adj25 low\$ back pain" or "mechanic\$ adj25 low\$ back pain" or "non-specific adj25 low\$ back pain" or backache or "back pain" or "intervertebral dis#" or "Intervertebral Dis# Degeneration" or "dis# degeneration" or "Intervertebral Dis# displacement" or "dis# displacement" or "simple back pain" or "low back syndrome" or "low back dysfunction" or non-specific back pain" or "lumbar pain" or lumbar vertebr\$").mp.
8. 6 or 7
9. Randomized Controlled Trial/
10. (((((randomied controlled trials or controlled clinical trials or random allocation or double-blind method or single-blind method or clinical trial).mp. or exp clinical trial/or clin\$.mp.) or (singl\$ or doubl\$).mp.) adj25 (blind\$ or mas\$).mp.) or placebo\$.mp. or placebo.mp. or random\$.mo. or research design.mo. or comparatice study.mo. or exp evaluation studies/ or follow-up studies.mp. or prospective studies.mp. or cross-over studies.mp. or (control\$ or prospective\$ or volunteer\$).mp. or parallel-stud\$.mp. or (parallel adj25 stud\$).mp. or (parallel.sh. adj25 trial\$).mp.
11. 9 or 10
12. 5 and 8 and 11
13. limit 12 to yr="2015-Current"

Appendix B: Data Extraction Tables

Table 2. Data Extraction Table										
Source	Enrichment Study? (1=Yes, 0=No)	Drug	Oral (1) vs Transdermal (2) vs Buccal (3)	Outcome Measure	N1	Mean1	SD1	N0	Mean0	SD0
Buynak et al 2010a	0	Oxycodone (53 mg/d) vs. placebo	1	NRS* Week 8	171	46.2	17.5	83	55.1	18.3
Buynak et al 2010a	0	Oxycodone (53 mg/d) vs. placebo	1	NRS; Week 12	145	46.2	16.4	83	55.1	18.3
Buynak et al 2010b	0	Tapendatol (313.2 mg/d) vs. placebo	1	NRS; Week 8	201	47.4	20.6	83	55.1	18.3
Buynak et al 2010b	0	Tapendatol (313.2 mg/d) vs. placebo	1	NRS; Week 12	183	46.4	19.9	83	55.1	18.3
Christoph et al, 2017a	1	Cebranopadol (200hg) vs placebo	1	NRS (0-10); Week 12	122	42	10.7	125	48.9	11.6
Christoph et al, 2017b	1	Cebranopadol (400hg) vs placebo	1	NRS (0-10); Week 12	120	39.7	10.6	125	48.9	11.6
Christoph et al, 2017c	1	Cebranopadol (600hg) vs placebo	1	NRS (0-10); Week 12	117	39.6	10.3	125	48.9	11.6
Christoph et al, 2017d	1	Tapenadol (200mg) vs placebo	1	NRS (0-10); Week 12	123	39.4	10.6	125	48.9	11.6
Cloutier et al, 2013	0	Oxycodone/naloxone (36.5mg/d)vs. placebo	1	VAS Week 8	32	48.6	23.1	31	55.9	25.4
Gimbel et al, 2016	1	Buccal buprenorphine (300-1800 uh/d) vs placebo	3	NRS Week 12	254	35	9	257	43.7	9.64
Gordon et al, 2010	0	Transdermal buprenorphine (29.8ug/h) vs. placebo	2	VAS Week 4	37	40.9	20.2	42	48	22.5
Hale et al 2005a	1	Oxymorphone (79.4 mg/d) vs placebo	1	VAS; Day 18	71	8.5	23.9	67	27.5	23.8
Hale et al 2005b	1	Oxycodone (155.0 mg.d) vs placebo	1	VAS; Day 18	75	6.5	23.8	67	27.5	23.8

Hale et al, 2010	1	Hydromorphone (26mg/d) vs. placebo	1	NRS (0-10); Week 8	133	31	19.4	133	41	18.8
Hale et al, 2010	1	Hydromorphone (26mg/d) vs. placebo	1	NRS Week 12	133	32	19.4	133	41	18.8
Hale et al, 2007	1	Oxymorphone (OPANA) (80.9 mg/d) vs placebo	1	VAS Day 10, <3months	49	31	16.8	18	55	15.4
Hale et al, 2015	1	Hydrocodone (48mg/d) vs. Placebo	1	NRS, Week 12	193	29.4	14.7	196	33	19.3
Katz et al, 2007	1	Oxymorphone ER (39.2 mg/d) vs. Placebo	1	VAS; Week 8	105	33	24.6	100	45	27.9
Katz et al, 2007	1	Oxymorphone ER (39.2 mg/d) vs. Placebo	1	VAS: Day 90	105	31	24.5	100	47	27.9
Katz et al, 2015	1	Xtampza (Extended-Release Oxycodone) (60mg/d) vs placebo	1	NRS Week 6	193	29.4	14.7	196	33	19.3
Katz et al, 2015	1	Xtampza (Extended-Release Oxycodone) (60mg/d) vs placebo	1	NRS Week 12	193	29.4	15.6	196	34	22
Peloso et al, 2004	0	Tramadol/paracetamol (158.0 mg/d) vs. placebo	1	VAS Day 91	167	47.4	15	169	62.9	15.5
Rauck et al, 2015	1	ALO-02 (70mg/d) vs placebo	1	NRS Week 12	145	33	18.6	133	39.7	19
Rauck et al, 2016	1	Buccal buprenorphine (600uh/day) vs placebo	3	NRS Week 12	211	34.8	9.18	209	40.4	10.3
Rauck et al, 2014	1	Hydrocodone (119.0 mg) vs. placebo	1	NRS Day 85	151	4.8	15.6	151	9.6	15.5
Ruoff et al, 2003	1	Tramadol/paracetamol (157.5 mg/d) vs. placebo	1	VAS Day 91	91	44.4	14.5	74	52.3	14.9
Schnitzer et al, 2000	1	Tramadol (242.0 mg/d) vs. placebo	1	VAS (0-100);Day 49	127	35	27.9	127	51	29.8
Steiner et al, 2011	1	Buprenorphine transdermal (15ug/h) vs placebo	2	NRS Week 12	257	38.1	19.4	283	43.9	21.2

Uberall et al, 2012	0	Transdermal buprenorphine (10 ug/h) vs placebo	2	NRS Week 4	107	35.8	18.4	110	37.6	18.3
Vorsanger et al, 2008a	1	Tramadol ER (300mg/d) vs placebo	1	VAS (0-100); Week 8	129	35	26	63	40	25.9
Vorsanger et al, 2008a	1	Tramadol ER (300mg) vs placebo	1	VAS (0-100); Week 12	129	38	26	63	39	25.9
Vorsanger et al, 2008b	1	Tramadol ER (200mg/d) vs placebo	1	VAS (0-100); Week 8	127	34	23.7	63	40	25.9
Vorsanger et al, 2008b	1	Tramadol ER (200mg/d) vs placebo	1	VAS (0-100); Week 12	127	33	23.7	63	39	25.9
Webster et al 2006a	0	Oxycodone (39mg/d) four times daily vs. placebo	1	NRS Week 1	206	39	25.3	101	54	28.7
Webster et al, 2006a	0	Oxycodone (39mg/d) four times daily vs. placebo	1	NRS Week 12	206	40	25.3	101	52	30.5
Webster et al, 2006b	0	Oxycodone (34.5mg/d) plus low dose naltrexone four times daily vs. placebo	1	NRS, Week 12	206	42	25.6	101	52	30.5
Webster et al, 2006b	0	Oxycodone (34.5mg/d) plus low dose naltrexone four times daily vs. placebo	1	NRS, Week 1	206	41	25.1	101	54	28.7
Webster et al, 2006c	0	Oxycodone (34.7 mg/d) plus low dose naltrexone twice daily vs. placebo	1	NRS, Week 1	206	42	25.5	101	54	28.7
Webster et al, 2006c	0	Oxycodone (34.7 mg/d) plus low dose naltrexone twice daily vs. placebo	1	NRS Week 12	206	43	25.5	101	52	30.5
Wen et al, 2015	1	Hydrocodone tablets (50mg/d) vs placebo	1	NRS Week 12	296	34	20.5	292	38.6	20.4

*All NRS pain scales converted to VAS pain scales using information provided by Hawker et al. (2011)¹⁷

Table 3. Adverse Events

Source	Constipation	Nausea	Dizziness	Headache	Vomiting	Pruritus	Somnolence	Dry-mouth	Withdrawal Syndrome	Other
Buynak et al, 2010	x	x	x	x	x	x	x	x		
Chu et al, 2012	x	x				x		x		
Cloutier et al, 2013	x	x	x		x		x	x		Respiratory Infection, Abdominal Pain
Cristoph et al, 2017	x	x	x	x	x		x		x	Hyperhidrosis, Fatigue
Gimbel et al, 2016	x	x	x	x	x		x		x	Diarrhea, fatigue, dyspnea
Gordon et al, 2010	x	x	x	x	x	x	x			Asthenia, Anorexia, Rash, Nervousness, Yawn, Insomnia
Hale et al, 2010	x	x		x	x		x			
Hale et al, 2005	x									
Hale et al, 2007	x	x		x	x	x	x		x	
Hale et al, 2015	x	x	x	x	x	x	x	x	x	
Katz et al, 2007	x	x								Anxiety, Erectile Dysfunction
Katz et al, 2015	x	x	x	x	x	x	x	x	x	Fatigue, UTI, Withdrawal Syndrome, Hypoacusis
Peloso et al, 2004	x	x	x	x	x		x	x		Anorexia, Sweating
Rauck et al, 2014	x	x	x	x	x	x	x	x	x	
Rauck et al, 2015	x	x		x	x				x	Brochitis, Depression, Suicide Attempt

HP407

88964

Rauck et al, 2016	x	x		x	x		x		x	Diarrhea, Respiratory Infection
Ruoff et al, 2003	x	x	x	x	x	x	x	x		Fatigue, Upper Respiratory Infection
Schiporst et al, 2013	Not Reported	Not Reported	Not Reported	Not Reported	Not Reported	Not Reported	Not Reported	Not Reported	Not Reported	Not Reported
Schnitzer et al, 2000		x		x	x		x			
Steiner et al, 2011		x	x	x	x					Gastrointestinal disorder, Dehydration
Uberall et al, 2012	x	x	x		x			x		Fatigue, Abdominal Discomfort, Loss of efficacy
Vorsanger et al, 2008	x	x	x		x	x	x		x	Fatigue, Insomnia
Webster et al, 2006	x	x	x		x	x	x			
Wen et al, 2015	x	x	x	x	x		x		x	

X* = reported in study

Table 4. Drop-out Rates				
Study	Number enrolled in initial study	Final Number of people actually contributing to outcomes	% of people actually contributing to outcomes	Overall Drop-Out Rate
Buynak et al 2010	334	133	39.8%	60.20%
Christoph et al, 2017	641	502	78.30%	21.70%
Chu et al 2012	69	48	69.50%	30.50%
Cloutier et al, 2013	83	63	75%	24%
Gimbel et al, 2016	815	353	43.30%	56.70%
Gordon et al, 2010	79	42	53%	46.80%
Hale et al 2005	330	139	42.10%	57.90%
Hale et al, 2010	325	66	20.30%	79.70%
Hale et al, 2007	143	67	46.80%	53.20%
Hale et al, 2015	391	294	75.10%	24.90%
Katz et al, 2007	226	71	31.40%	68.60%
Katz et al, 2015	389	222	57.10%	42.90%
Peloso et al, 2004	338	146	43.10%	56.90%
Rauck et al, 2015	410	281	68.50%	31.50%
Rauck et al, 2016	749	420	56%	44%
Rauck et al, 2014	510	302	59%	40.70%
Ruoff et al, 2003	322	165	51.20%	48.80%
Schiphorst, 2013	Not reported	Not reported	Not reported	Not Reported
Schnitzer et al, 2000	380	146	38.40%	61.60%
Steiner et al, 2011	1027	540	53%	47%
Uberall et al, 2012	363	276	76%	24%
Vorsanger et al, 2008	386	155	40.20%	59.80%
Webster et al 2006	719	328	45.60%	54.40%
Wen et al, 2015	598	439	75%	25%

Table 5. DSM-5 Definition of opioid misuse or defined by the authors

Source	<i>Opioids are often taken in larger amounts or over a longer period than was intended</i>	<i>Persistent desire or unsuccessful efforts to cut down use</i>	<i>Time is spent in activities necessary to obtain the opioid</i>	<i>Craving, or strong desire or urge to use opioids</i>	<i>Recurrent use resulting in a failure to fulfill major role obligations</i>	<i>Continued use despite having persistent or recurrent social or interpersonal problems</i>	<i>Social or recreational activities are given up or reduced</i>	<i>Recurrent use in situations physically hazardous</i>	<i>Tolerance, defined by need for increased amounts of opioids to achieve</i>	<i>Withdrawal as manifested by the characteristic opioid withdrawal syndrome</i>
Cloutier et al, 2013										"No reports of withdrawal symptoms by the investigators at any time during the study"
Cristoph et al, 2017										"Data from the SOWS showed that cases of mild withdrawal symptoms were reported at similar low frequencies in the 3 cebranopadol arms (4.6%-6.5%), and reported more often in the tapentadol PR arm (9.9%) compared to the cebranopadol arms."
Gimbel et al, 2016										"The incidence of drug withdrawal during this period was low at 1.2%"
Hale et al, 2007										"ARS scores remained low, indicating that most patients did not experience symptoms of opiate withdrawal. This
Hale et al, 2015	"The majority (=75%) of patients randomized to hydrocodone ER (n=104; 71%) and			"three reported theft of rescue medication (hydrocodone/acetam						"the majority of patients classified their opioid withdrawal symptoms as normal...Mean COWS scores and the proportions of patients in each category of withdrawal
Rauck et al, 2014	"Discontinuation for excessive use of rescue medication"									"Discontinuation for opioid withdrawal occurred in 4.6% of subjects in the placebo group and 0% of subjects in the HC-ER group" following enrichment phase

Rauck et al, 2015										"Eight patients experience adverse events characteristic of 'withdrawal syndrome' during the course of the study with most events occurring because of opioid cessation"... "Twelve patients (3.2%) had a maximum COWS score between 5 and 12, indicating mild withdrawal symptoms."
Rauck et al, 2016										"Drug withdrawal syndrome was reported by fewer than 3% of patients randomized to placebo, and only one patient discontinued the study because of opioid withdrawal."
Schnitzer et al, 2000										"An examination of the adverse event profile of the patient who received placebo at the start of the double-blind treatment gave no indication of opioid withdrawal syndrome."
Steiner et al, 2011	One patient receiving BTDS 20 experienced "immediate-release oxycodone abuse"									
Uberall et al, 2012									Reported loss of efficacy, Table 5	

Vorsanger et al, 2008										"The finding of a significantly higher proportion of patients with opioid-type withdrawal symptoms in the tramadol ER groups compared to the placebo group one week after abrupt discontinuation of treatment in this study is consistent with previous reports from studies of tramadol-containing products, including tramadol ER."
Wen et al, 2015										"During the first 2 weeks of the double-blind period, four patients (1%) in the placebo group experienced AEs of opioid withdrawal as defined by SMQ subcategory of drug withdrawal and three patients (1%) in the placebo group experienced AEs of opioid withdrawal as defined by DSM-IV. "

***Only studies that report assessing signs of opioid misuse/abuse/withdrawal as defined by the DSM-5 or by the authors are shown here. All other studies did not report about misuse/abuse/withdrawal.**

Appendix C: Risk of Bias

Table 6. Summary of Risk of Bias Assessment

Study	Sequence Generation Description	Sequence Generation Judgement	Allocation Concealment Description	Allocation Concealment Judgement	Blinding (Personnel & Participants)Description	Blinding (Personnel & Participants) Judgement	Blinding (Outcome Assessment) Description	Blinding (Outcome Assessment) Judgement	Incomplete Outcome Data Description	Incomplete Outcome Data Judgement	Selective Outcome Reporting	Selective Outcome Reporting Judgement
Christoph et al (2017)	"Block randomisation with stratification factors;Randomization of patients took place at the Baseline Visit (start of the treatment period), based on computer-generated randomization lists provided by an external supplier and was implemented using an interactive response technology system ... block randomization with stratification factors"	Low	"Computer-generated randomisation lists provided by an external supplier and implemented using an interactive response technology system"	Low	"double-blind, double-dummy"	Low	"double-blind, double-dummy"	Low	Attrition and exclusions reported, with reasons. Attrition: placebo 25/126; cebranopadol 200ug 61/131, cebranopadol 400ug 67/128, cebranopadol 600ug 76/130, tapentadol 49/126. LTF (1 placebo, 2 cebranopadol 200ug, 1 tapentadol 200mg) similar across treatment groups. "Other" similar across treatment arms	High	Completed primary endpoints as per protocol	Low
Cloutier et al (2013)	"A computer-generated random allocation listing of patient numbers was generated by the biostatistician and patients entering the study were centrally allocated to the associated treatment condition."	Low	"A computer-generated random allocation listing of patient numbers was generated by the biostatistician and patients entering the study were centrally allocated to the associated treatment condition."	Low	"Blinding was achieved using the double dummy technique. The patients, investigator and all clinical research staff were blinded"	Low	"Blinding was achieved using the double dummy technique. The patients, investigator and all clinical research staff were blinded."	Low	20/83 did not complete the protocol (cross-over design) - fairly even split between placebo and treatment arms. LTF 2 in first placebo group, remainder due to adverse events (7 treatment, 5 placebo), consent withdrawal and protocol violations	Low	Unable to see whether protocol amended after trial	Unclear
Gimbel et al (2016)	"Randomisation by stratification... Randomisation codes were generated by sponsor/designee to ensure appropriate numbers in each treatment arm."	Low	"Randomisation codes were generated by sponsor/designee... The randomisation codes were stored in a secure electronic format within IXRS system."	Low	"All patients, investigators and staff employed by investigators/sponsor were blinded to treatment assignments"	Low	"All patients, investigators and staff employed by investigators/sponsor were blinded to treatment assignments"	Low	n=254 BBUP (48 discontinued - 19 lack of efficacy; 1 LTF; 7 other) 243 analyzed for primary endpoint n=257 placebo (110 discontinued - 61 lack of efficacy; 5 LTF; 5 other) 248 analyzed for primary endpoint. Reasons given for discontinuation	High	Primary outcome data reported. Some secondary outcome data not reported in results section: 3, 7, 8	Low
Gordon et al (2010)	"Blocked randomisation... The generation of the random code was facilitated by the use of PROC PLAN in SAS version 6.12"	Low	"The generation of the random code was facilitated by the use of PROC PLAN in SAS version 6.12"	Low	"Double Blinded"	Low	"Double blinded"	Low	28/79 patients withdrawn, 19 treatment arm vs 9 placebo arm (cross-over study). mostly due to AE (n=23: 5 placebo, 18 treatment). none LTF. all reasons given.	High	Unable to see when primary/secondary outcomes altered on protocol after trial	Unclear
Hale et al (2005)	"A randomization code was generated by INC Research... the study treatment was blocked by site to help ensure equal distribution of patients to treatment groups within a site"	Low	"Unique 4-digit numbers were assigned to each study kit and distributed in numeric order at the site, starting with the lowest numbered kit." "To break the randomization code in case of an emergency, the study medication bottles had a scratch-off label."	Low	double-blind dose-titration and treatment phases "All participants, investigators, and staff remained blinded to treatment during the study." "The study medications were made indistinguishable by overencapsulating the tablets with gelatin and were color coded to differentiate the dose strengths."	Low	"All participants, investigators, and staff remained blinded to treatment during the study." "The study medications were made indistinguishable by overencapsulating the tablets with gelatin and were color coded to differentiate the dose strengths."	Low	Discontinued: 22/80 oxymorphone ER, 53/75 placebo, 21/80 oxycodone CR 2 LTF - 1 oxy ER and 1 placebo, largest cause of discontinuation was lack of efficacy (26 oxy ER, 53 placebo, 21 oxy CR). all specified reasons bar the 2 LTF and 1 'other' (oxy CR)	High	Unable to locate protocol	Unclear
Hale et al (2007)	"Randomized"	Unclear	-	Unclear	"Double blinded", The open label phase could be seen as biasing the study by affecting the patients' treatment assignment, but not dosage, during study"	Low	"Double Blind", The open label phase	Low	Discontinued: 21/70 treatment, 55/73 placebo. most due to lack of efficacy (placebo n=39; treatment n=12)	High	Unable to locate original protocol	Unclear
Hale et al (2015)	"Patients were randomly assigned to treatment 1:1 through a qualified randomization service provider using an interactive response technology system"	Low	"qualified randomization service provider using an interactive response technology system"	Low	"Patients and investigators remained blinded to treatment assignment, but not dosage, during study"	Low	"Patients and investigators remained blinded to treatment assignment, but not dosage, during study"	Low	Discontinued: 35 treatment group (18%), 38 placebo (21%). LTF: 2 treatment group, 2 placebo Other reasons for discontinuation listed, bar 1 'other' for treatment group > clear reasons given for attrition, similar between treatment and placebo arms	Low	Primary and main secondary outcomes reported as per protocol; Additional secondary outcomes added to protocol after trial - unrelated to pain assessment so limited bias for current study	Low
Katz (2015)	"Patients were randomized using a block randomization scheme in a 1:1 ratio, randomization was stratified by previous opioid use. Patients were assigned a unique randomization number by an interactive web response system; an external-to-study, unblinded biostatistician prepared randomization code"	Low	"Patients were assigned a unique randomization number by an interactive web response system; an external-to-study, unblinded biostatistician prepared randomization code".	Low	"Double blind"	Low	"Double Blind"	Low	"Discontinuation reasons and numbers given clearly for both treatment groups; n=71 (36.8%) for Xtampza ER - 8 lack of efficacy, 3 LTF, 4 other n=96 (49.0%) for placebo - 34 lack of efficacy, 7 LTF". Reasons given, similar numbers between treatment groups.	High	Primary outcome reported as per protocol. Secondary outcome measures added to protocol after trial - but this does not affect pain results to limited bias for current study	Low

Peloso et al (2004)	"Patients were randomized in a 1:1 fashion to tramadol/acetaminophen or placebo, using a centrally prepared randomization scheme carried out in blocks of 8."	Low	"centrally prepared"	Low	"Double blind... Blinding was accomplished by packaging study medications to appear identical for all treatment groups and by concealing the composition and strength of the test preparations under a label mask."	Low	"Double blind... Blinding was accomplished by packaging study medications to appear identical for all treatment groups and by concealing the composition and strength of the test preparations under a label mask."	Low	Withdrawals: 81/167 treatment, 110/169 placebo, none reported as LTF, 9 withdrew due to 'other' - 1 treatment, 1 placebo. Most withdrawals due to (1) insufficient pain relief, treatment n=30; placebo n=82 (2) adverse event, treatment n=47, placebo	High	Not able to locate clinical trial protocol	Unclear
Rauck et al (2014)	"Randomized"	Unclear	-	Unclear	"Double blind", "matching placebo", The open label phase could be seen as biasing the study by affecting the patients' ability to determine if they are receiving active treatment or placebo during the double blind phase	Low	"Double blind", "matching placebo", The open label phase could be seen as biasing the study by affecting the patients' ability to determine if they are receiving active treatment or placebo during the double blind phase	Low	151 treatment arm: 27 discontinued - 1 LTF, remaining reasons listed (highest proportion due to lack of efficacy, n=14) 151 placebo arm: 92 discontinued - 1 'other', remaining reasons listed (highest proportion due to lack of efficacy, n=62) remaining reasons balanced between the 2 treatment arms	High	Primary/Secondary outcomes as per protocol	Low
Rauck et al (2015)	"Patients were randomized in a 1:1 ratio"	Unclear	-	Unclear	"Double blind"	Low	"double blind"	Low	53/134 placebo discontinued 40/147 ALO-02 discontinued	High	Primary and secondary outcomes	Low
Rauck et al (2016)	"Patients were randomized 1:1"	Unclear	-	Unclear	"Double blind"	Low	"Double blind"	Low	n=209 treatment arm - 54 discontinued, 8 due to lack of efficacy and 7 due to other. 4 LTF n=211 placebo - 58 discontinued, 23 due to lack of efficacy	High	Primary outcome reported as per protocol. Secondary	Low
Ruoff et al (2003)	"Randomization was performed using SAS version 8 (SAS Institute Inc., Cary, North Carolina). Block randomization was by site."	Low	"Randomization was performed using SAS version 8"	Low	"Patients, investigators, clinical staff, and study monitors remained blinded to treatment assignments until therapy was complete and the database was finalized."	Low	"Patients, investigators, clinical staff, and study monitors remained blinded to treatment assignments until therapy was complete and the database was finalized."	Low	Tramadol: 162 70 withdrew, most due to insufficient pain relief (n=31) and adverse event (n=30). 1 'other'; none LTF Placebo: 160 83 withdrew, most due to insufficient pain relief (n=59) and adverse event (n=9). 8 'other'; none LTF	High	Unable to locate protocol	Unclear
Schiphorst et al (2014)	"Randomization was done in blocks of 10 for one of the two conditions (placebo or experimental) [according to a] randomization scheme."	Low	For the randomization boxes: "Only the hospital pharmacist had access to the key. The code was broken when writing of the manuscript was in its final stage."	Low	"Clinician/researcher, tester, patients were blinded to the treatment provided during the study, data-analysis, and writing process of the manuscript until the last version (triple blinded)." "Treatment and placebo capsules were delivered blinded"	Low	"Clinician/researcher, tester, patients were blinded to the treatment provided during the study, data-analysis, and writing process of the manuscript until the last version (triple blinded)." "Treatment and placebo capsules were delivered blinded"	Low	n=25 placebo: 3 discontinued, none LTF, all reasons given (none related to study) n=25 treatment: 1 LTF, 3 discontinued, all reasons given (n=2 side-effects) low numbers of discontinuation/even split between treatment arms	Low	Not able to locate trial protocol	Unclear
Schnitzer et al (2000)	"computer generated random numbers were used to ensure that any given patient would be assigned randomly to one of the 2 treatment groups"	Low	"computer generated random numbers"	Low	"Identical appearing capsules of tramadol HCl and placebo were prepared" "double blind","The open label phase could be seen as biasing the study by affecting the patients' ability to determine if they are receiving active treatment or placebo during the double blind phase"	Low	"Identical appearing capsules of tramadol HCl and placebo were prepared" "double blind","The open label phase could be seen as biasing the study by affecting the patients' ability to determine if they are receiving active treatment or placebo during the double blind phase"	Low	Discontinuation: 36/127 tramadol - inadequate pain relief largest reason (n=25), adverse event (n=5), other (n=1), no LTF 72/127 placebo - inadequate pain relief largest reason (n=25), adverse event (n=5), other (n=1), no LTF	High	unable to locate original protocol	Unclear
Steiner et al (2011)	"patients... were randomized in a 1:1 ratio"	Unclear	-	Unclear	open-label run-in period can disrupt patient blinding (patients may be able to guess which treatment they had)	Low	"double blind", open-label run-in period can disrupt patient blinding (patients may be able to guess which treatment they had)	Low	Discontinuation: Placebo = 84/283 -> lack of effect largest reason (n=36), adverse event (n=20), 'other' (n=28)	High	Primary and secondary outcomes as reported in	Low
Uberall et al (2012)	"patients were randomly assigned in a 1:1:1 ratio... Randomization of patients to treatment was based on a computer-generated allocation list with a block-size of six, and stratified by study site."	Low	"Computer-generated allocation list with a block-size of six, and stratified by study site."	Low	"matching placebo" "double blind"	Low	"matching placebo" "double blind"	Low	Discontinuation: Placebo - 26/122: Adverse event (n=4), lack of efficacy (n=2) LTF (n=2) Flupirtine - 28/123: Adverse event (n=4), lack of efficacy (n=2), LTF (n=4) Tramadol - 33/118: Adverse event (n=14), lack of efficacy (n=1) LTF (n=2)	High	Protocol does not list whether changes to primary and secondary outcomes were made after start of trial	Unclear
Vorsanger et al (2008)	"patients... were randomized in a 1:1:1 ratio to receive double-blind tramadol 300 mg, 200 mg, or placebo. A computer-generated randomization schedule was prepared and patients were sequentially assigned to one of three treatments according to the randomization list."	Low	"A computer-generated randomization schedule was prepared and patients were sequentially assigned to one of three treatments according to the randomization list."	Low	"open-label run-in" - may unblind participants if they can guess their treatment status tablets "were identical in appearance and texture to maintain the double-blind"	Low	"open-label run-in" - may unblind participants if they can guess their treatment status tablets "were identical in appearance and texture to maintain the double-blind"	Low	Discontinuation: 42/128 Tramadol 300 - Adverse events (n=13), lack of efficacy (n=13) 42/129 Tramadol 200 - Adverse events (n=13), lack of efficacy (n=11) 61/129 placebo -Adverse events (n=18), lack of efficacy (n=21) >> "statistically non-significant changes in numbers between treatment arms"	High	Primary and secondary outcomes as per protocol	Low
Webster et al (2006)	"Qualifying patients were randomized via a central call-in system to 1 of 4 treatments in a 1:2:2:2 ratio of placebo, oxycodone qid, oxytrex qid"	Low	"Central Call-in System"	Low	"double blind" "All study medications were identical in appearance, and patients, site personnel, and study monitors were blinded to treatment assignments."	Low	"double blind" "All study medications were identical in appearance, and patients, site personnel, and study monitors were blinded to treatment"	Low	Discontinuation: Placebo 59/101 - inadequate pain relief (n=40), adverse events (n=5), 'other' (n=6) Oxycodone 105/206 - inadequate pain relief (n=15), adverse events (n=49), 'other' (n=27) Oxytrex QID 119/206 - inadequate pain relief (n=22), adverse events (n=45), 'other' (n=36) Oxytrex BID 108/206 - inadequate pain relief (n=19), adverse events (n=63), 'other' (n=17)	High	unable to locate study protocol	Unclear
Wen et al (2015)	"Randomization was stratified by HYD dose at end of run-in period and by opioid status; patients were randomized in a 1:1 ratio"	Unclear	-	Unclear	"During the first 2 weeks of the double-blind period, patients randomized to placebo had their HYD doses tapered in a blinded manner, with a reduction in dose of approximately 25 -- 50% every 3 -- 4 days."	Low	"During the first 2 weeks of the double-blind period, patients randomized to placebo had their HYD doses tapered in a blinded manner, with a reduction in	Low	n=149 (25%) discontinued, 82 (28%) placebo and 67 (23%) HYD. LTF 3 (1%) placebo and 5 (2%) HYD. lack of effect 44 (15%) placebo and 16 (5%) HYD. full details given for reasons for discontinuing, numbers similar between treatment groups and LTF numbers small.	High	Primary and secondary outcomes reported as in protocol	Low

Appendix C (continued):

Risk of Bias PEDro ratings	Total Score
Buynak et al, 2010a	8
Chu et al, 2012	8
Hale et al, 2010	8
Katz et al, 2007	8

Figure 6. Risk of bias from Shaheed et al. (2016) PEDro scale ratings on inaccessible articles.

Appendix D: Meta-regressions

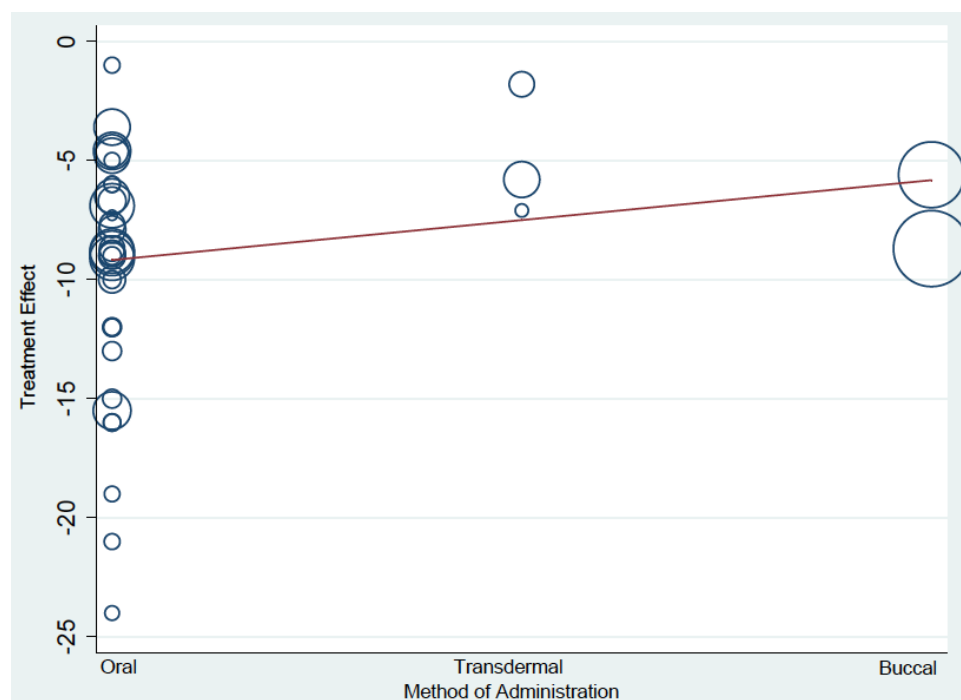


Figure 7. Meta-regression of method of administration on treatment effect

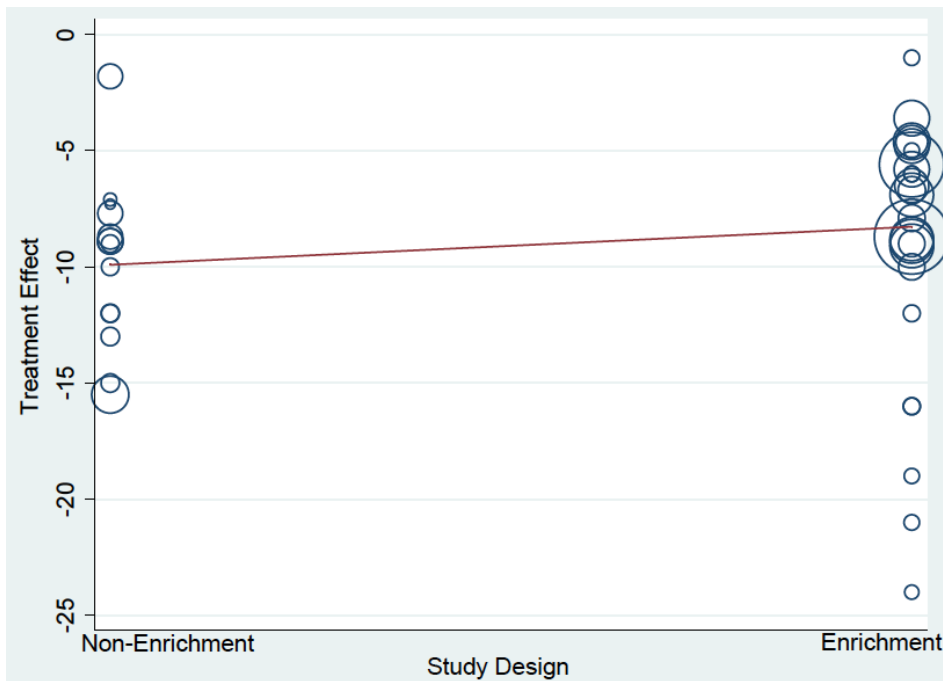


Figure 8. Meta-regression of study design on treatment effect

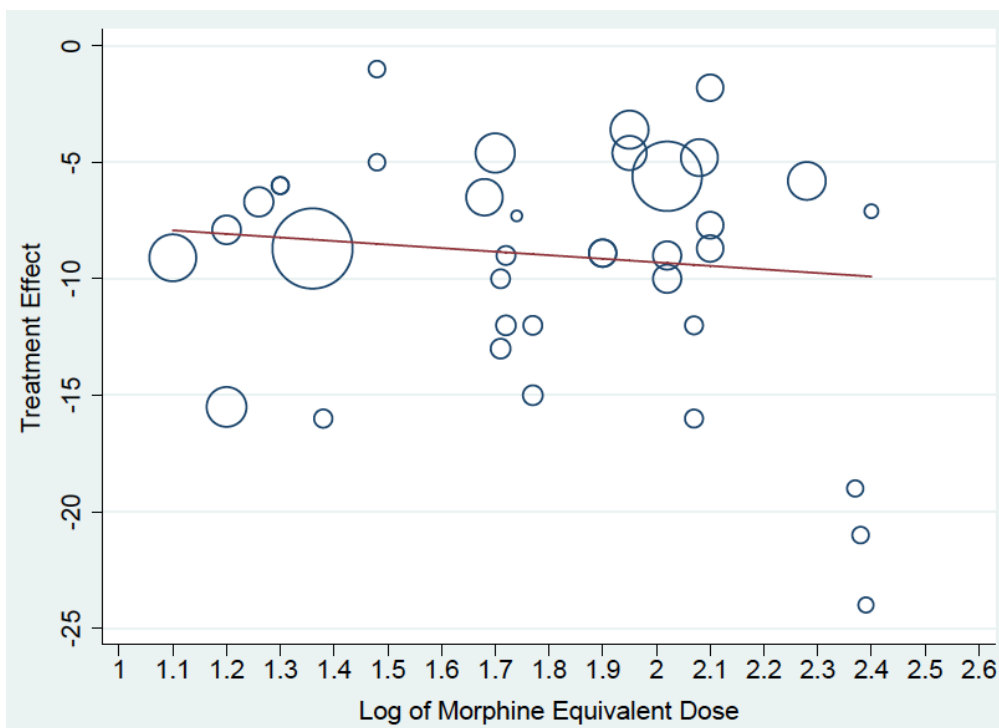


Figure 9. Meta-regression of log-transformed morphine equivalent dose on treatment effect